

Summative Assessment 2: Data Analysis Project Brief



Summative Assessment 2: Data Analysis and modelling project

Key Information

This assessment evaluates your practical skills in programming for data science across Weeks 1-6 of the module.

Component	Marks
Part 1 – Data Analysis Project	32 marks
Part 2 – Machine Learning Pipeline	32 marks

Process Log	12 marks
Report Quality	4 marks
Total	80 marks

Release date

Assessment brief available from Week 1. The fixed data set, topic list and project Jupyter Notebook template will be released on **Friday 18 September 2026, 14:00 UTC+1**.

Deadline

Monday 26 October 2026, 14:00 UTC (London)

Purpose

The purpose of this assessment is to develop your skills in devising and presenting a Data Science project.

Assessment Task

Part 1 – Data Analysis Project

For Part 1, you must choose your own dataset, and it must be publicly available and suitable for exploratory data analysis. Recommended sources include:

- [Kaggle \(https://www.kaggle.com/datasets\)](https://www.kaggle.com/datasets)
- [UCI Machine Learning Repository \(https://archive.ics.uci.edu/\)](https://archive.ics.uci.edu/)
- [scikit-learn sample datasets \(https://scikit-learn.org/stable/datasets.html\)](https://scikit-learn.org/stable/datasets.html)
- Open data portals and APIs (e.g. government, WHO, World Bank)

You may use scikit-learn only to load a built-in dataset (e.g. `sklearn.datasets`). You must not use the library for any other part of your analysis in this section: no preprocessing, transformations, or statistical functions from scikit-learn. Part 1 should otherwise rely on core Python, pandas, and a visualisation library only; the full scikit-learn toolkit is reserved for Part 2.

Part 2 – Machine Learning Pipeline

For Part 2 you will work with a fixed dataset provided by the module team. This dataset supports a binary classification problem, contains a mixture of numerical and categorical features, and includes a clearly defined target variable. The exact dataset and filename for the current run will be provided in the notebook template.

Part 1 – Data Analysis Project (40 marks)

Component	Marks
-----------	-------

Section 1.1 – Project Aim and Dataset description and source	10 Marks
Section 1.2 – Code quality and workflow	12 marks
Section 1.3 – Objectives analysis	6 Marks
Section 1.4 – Insights and reflection	4 Marks
Total	32 Marks

Section 1.1 – Project Aim and Dataset Description (10 marks)

State your project aim and define two specific, measurable objectives that are relevant to your chosen dataset and domain. Write a description of your dataset covering its source, contents, quality, and suitability for your intended analysis.

Expected length: approximately 400–500 words

Section 1.2 – Exploratory Data Analysis and Code (12 marks)

Write well-structured Python code to explore your dataset. Your code should be logically organised, clearly commented, and easy to follow. Use descriptive variable names and include comments that explain why each step is being performed. After each block of code, include a brief markdown cell explaining what the output shows and what it means for your analysis.

The combined code for **Sections 1.2 and 1.3 must not exceed 250 non-empty lines**. The written commentary for this section 1.2 must not exceed **600 words**. These are maximum limits rather than recommended targets; quality, relevance and clarity matter more than quantity.

Section 1.3 – Objectives Analysis (6 marks)

For each objective, present your analysis and findings supported by at least one visualisation. You must:

- Quote at least one specific numerical value visible in the visualisation or output (for example, a peak value, a proportion, a correlation coefficient, or an outlier value).
- Explain what that value means in the context of the domain, not just in terms of the data.

Expected length: approximately 250–325 words per objective, plus visualisations.

Section 1.4 – Insights and Reflections (4 marks)

Write a summary of your analysis covering:

- Your main findings.
- Anything unexpected you encountered.
- The limitations of your work.

Expected length: approximately 200–250 words.

Part 2 – Machine Learning Pipeline (32 marks)

Component	Marks
Section 2.1 – Preprocessing and pipelines	10 marks
Section 2.2 – Baseline model training	4 marks
Section 2.3 – Cross-validation	6 marks
Section 2.4 – Hyperparameter tuning	6 marks
Section 2.5 – Final model evaluation and justification	6 marks
Total	32 Marks

Section 2.1 – Preprocessing and Pipelines (10 marks)

Prepare the dataset and build a scikit-learn preprocessing pipeline. After the code, write a reflection of approximately 200–250 words justifying the preprocessing decisions you made for this dataset.

Section 2.2 – Baseline Model Training and Evaluation (4 marks)

Train two classification models as baselines. For each model, build a complete pipeline that incorporates your preprocessor. After the code, justify your choice of classifiers. Expected length: approximately 100–150 words.

Section 2.3 – Cross-Validation (6 marks)

Evaluate both baseline pipelines using cross-validation. Report at least one metric beyond accuracy, including mean and standard deviation across folds.

After the code, write a reflection of approximately 200–250 words comparing the two models and discussing what the results indicate about performance and stability. Identify which model you would take forward and justify your choice.

Section 2.4 – Hyperparameter Tuning (6 marks)

Select one model to tune. Apply a hyperparameter search to the full pipeline and report the best configuration found. Compare the tuned performance with your baseline results from Section 2.3.

After the code, write a reflection of approximately 200–250 words justifying your choices and evaluating whether tuning produced a meaningful improvement.

Section 2.5 – Final Model Evaluation and Justification (6 marks)

Evaluate your final model on the held-out test set. Report appropriate metrics, including precision, recall and F1, and display a confusion matrix with labelled axes. Compare the test performance with your earlier cross-validation results.

After the code, write a justification of approximately 250-300 words interpreting the results, discussing how well the model generalises, and providing a reasoned recommendation on whether the model is fit for purpose or where further improvement is needed.

Process Log (12 marks)

Submit a process log of no more than 400 words as part of your notebook, in the cell provided at the end of the template.

Your process log must address all three of the following:

- **Dead ends:** Describe at least two things you tried from parts 1 and 2 that did not work during your analysis or modelling. Be specific.
- **Surprising finding:** Identify one finding from your analysis that you did not expect. Reference the specific value or output.
- **What you would do differently:** If you were starting again with more time, what is the one thing you would change? Ground your answer in your own results.

Report Quality (4 marks)

These marks assess the overall presentation and professionalism of your submitted notebook, independent of the technical content assessed elsewhere. Specifically:

- All template placeholder and instructional text has been removed, leaving only your own work.
- The notebook is well-organised, with outputs visible and section headings intact.
- Writing throughout is clear, grammatically correct, and free of significant typos.
- The notebook reads as a coherent, professional piece of work rather than a set of disconnected answers.

General Marking Guidance

Marks are awarded for the quality of your reasoning and the clarity of your code and writing, not for achieving high model accuracy. A well-justified decision based on modest results will score more highly than a high-performing model with no explanation.

Across both parts, marks are awarded for:

- Correctness and appropriateness of your code.
- Quality of written interpretation grounded in specific values from your outputs.
- Clarity of code structure, comments, and variable naming.
- Evidence of critical thinking in your decisions and reflections.
- Consistency between your code, outputs, written commentary.

Acknowledgement of AI use

If you have used Gen AI as a functional tool to assist you as you were creating any academic work, you **must** acknowledge your use of AI (<https://generative-ai.leeds.ac.uk/ai-and-assessments/acknowledging-use-of-ai/>) in your submission.

Acknowledgements should include the name of the system and a brief description of how it has been used.

For example: 'I acknowledge the use of Microsoft Copilot in the preparation of this assignment. The tool was used to define concepts, give feedback, and support proofreading.'

Word and code limits

Before submitting, delete all template placeholder text and retain only your own responses. The word-count tool at the bottom of the notebook counts all markdown text, including any placeholder instructions that have not been removed.

The following maximum limits apply:

- **Maximum 600 lines of code** (combined across Part 1 and Part 2)
- **Maximum 3800 markdown words** (combined across Part 1, Part 2, and the Process Log)

These are maximum limits, not recommended targets. Clear, concise and well-structured solutions are encouraged, and students are not expected to use the full allowance. Code comments are not included in the markdown word count but may be considered when assessing code quality and conciseness.

Academic Integrity

As a student at the University of Leeds, you need to adhere to the University's regulations on academic misconduct. All work you submit for assessment is expected to be your own independent work. Where you have worked with others or where you present other people's work as part of your assessment, you must fully acknowledge this.

All students should ensure that they are aware of the principles of Academic Integrity. For more information or if you are unsure about academic integrity, please read the University of Leeds' [guidance on Academic Integrity](https://library.leeds.ac.uk/info/1401/academic_skills/46/academic_integrity_and_plagiarism) (https://library.leeds.ac.uk/info/1401/academic_skills/46/academic_integrity_and_plagiarism). Please note that by submitting this piece of work you are agreeing to the University's [Declaration of Academic Integrity](https://secretariat.leeds.ac.uk/student-cases/academic-misconduct/declaration-of-academic-integrity/) (<https://secretariat.leeds.ac.uk/student-cases/academic-misconduct/declaration-of-academic-integrity/>).

Guidance on the use of assistance

You are responsible for the academic integrity of the work you submit, and the work must be an accurate reflection of your own learning, understanding and skills. You can use assistance in a supportive capacity (i.e. support with work prior to submission for assessment). You are responsible for the process of drafting and refining the work you submit, including generating content, revising, editing and proofreading. It is important to be aware of the limitations of any assistance you use, and you are expected to engage critically with any outputs of assistance.

For more information or if you are unsure about the use of assistance, please read the University of Leeds' [Academic Integrity and Assistance Policy](https://students.leeds.ac.uk/academic-integrity/doc/assistance-coursework-taught-students) (<https://students.leeds.ac.uk/academic-integrity/doc/assistance-coursework-taught-students>).

Use of technologies in this assessment

The use of assistance, including Generative AI, is permitted for the preparation of this assessment.

As a general principle, you can use Gen AI to help you learn but cannot use AI to generate or falsify work. Remember, if you use Gen AI, you **must** acknowledge it (<https://generative-ai.leeds.ac.uk/ai-and-assessments/acknowledging-use-of-ai/>) in your submission.

In this assessment, AI tools **can** be used in an assistive role to:

- Ask questions about concepts or techniques you are trying to understand, for example asking what the difference between two preprocessing strategies is or how a particular function behaves;
- Test and debug code you wrote yourself, for example pasting an error message and asking what it means;
- Support proof-reading of written sections you have drafted yourself.

Gen AI **cannot be used in any other way** to produce content that you submit. In particular, it should not be used to:

- generate any portion of your notebook code, even as a starting point that you then modify;
- generate, rewrite, or substantially rephrase any portion of your written reflection or analytical justifications;
- make analytical decisions on your behalf, for example asking AI which preprocessing strategy, model, or hyperparameter setting to use, and then presenting that recommendation as your own reasoning;
- produce any content that you submit as evidence of your own understanding.

The written reflection

- Your justifications must refer specifically to characteristics of the dataset you observed during your own analysis, including the distributions, missing value patterns, class balance, and model performance figures you found. A reflection that could have been written without opening the dataset will not receive marks for justification, regardless of whether AI was involved in producing it.

Remember that you may be asked to demonstrate your authorship of this assessment. Save your working drafts, any AI prompts you used, and the outputs those prompts produced. If there is a question about your submission, you may be asked to explain your decisions and your process.

If you have any questions about the appropriate use of Gen AI in your assessed work, please review the University's AI and Assessments (<https://generative-ai.leeds.ac.uk/ai-and-assessments/>) guidance, and if this does not answer your question(s), contact your Module Leader.

Submission

Before you submit:

- Ensure the notebook runs from top to bottom without errors using Kernel → Restart & Run All.
- Do not submit extra files unless explicitly requested.
- Ensure your Student ID is entered in the designated cell at the top of the notebook.
- Do not delete or modify any blue section headings. These are used to structure your submission and must remain in place exactly as provided.

Submit your report by the due date and time using the submission link in **'Submit My Work'**.

You need to submit your report as a Jupyter notebook (.ipynb) file. The title of your notebook should be named **exactly as follows** in order to be accepted by the autograder:

P4DS_A2_Data_Analysis_Project.ipynb

Use the **Select File(s)** button to select the correct file(s) that you want to upload from your device.

When the upload is complete, the **Submit & View** Submission button will become active; select it to complete your submission.

For this assignment type, you **will not receive email confirmation** of your submission.

Extensions and Late Submission Guidance

If you will not be submitting your work before the deadline, you may submit to the Extensions and Late Submissions link (where permitted). You will incur a late penalty in accordance with the University of Leeds Assessment Policy and Procedures unless you've been granted a coursework extension. If the assessed work is submitted over 14 calendar days after the original submission deadline for any reason (including granted extensions), the submission will be deemed to have failed for non-submission (a day being a single 24-hour period).

Submissions should only be made to the Extensions and Late Submissions link if you fail to submit by the assessment deadline or have been granted a coursework extension. If you have already made a submission to the on-time submission link, any work resubmitted to the Extensions and Late Submissions link will not be marked. If you have already made a submission to the on-time submission link, any work resubmitted to the Extensions and Late Submissions link will not be marked.

If you have additional needs/unexpected circumstances which mean you need an extension to the deadline, please read the [University of Leeds Mitigating Circumstances Guidance](https://students.leeds.ac.uk/assessments/doc/mitigating-circumstances) (<https://students.leeds.ac.uk/assessments/doc/mitigating-circumstances>) and apply for a coursework extension via the [University Mitigating Circumstances online form](https://forms.office.com/pages/responsepage.aspx?id=qO3qvR3IzkWGPIIypTW3yy-0xsYxzEhHtZIB-qRq7GJUM09XWVFFUVU0UzVQQzM4WEcwSTA2WUNIOSQIQCN0PWcu) (<https://forms.office.com/pages/responsepage.aspx?id=qO3qvR3IzkWGPIIypTW3yy-0xsYxzEhHtZIB-qRq7GJUM09XWVFFUVU0UzVQQzM4WEcwSTA2WUNIOSQIQCN0PWcu>) in advance of the assessment deadline.

Marking

This is a compulsory assessment for OCOM5100M. It is worth **80%** of your overall module grade.

Feedback

You will receive your provisional mark and feedback via Minerva, in the Gradebook tab, accessed at the top of the Module Content page.

We will notify you by email when your feedback is available. You will usually receive feedback within 15 working days of the assignment deadline. This feedback may take the form of a written or recorded audio comment.

Next steps

Now continue to **OCOM5100M Data Analysis Project Topics**.